WORKED SOLUTION

STEP II 1992, Question 11

Mechanics • Projectile Motion

STEP II 1992, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q11

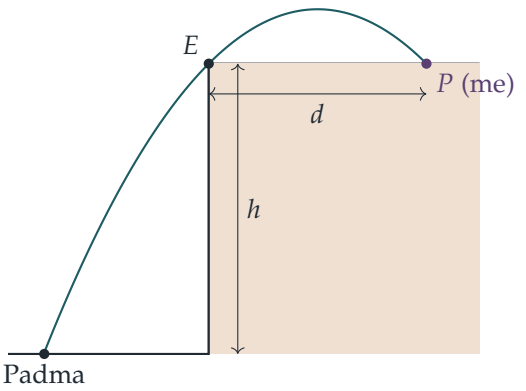
The key idea

The awkward part of the throw is not getting the wallet up to height h but getting it across the top of the cliff: it must be at height at least h all the way from the cliff edge to me, a horizontal distance of at least d. So look at the motion only from the moment the wallet first reaches height h. From then on it is an ordinary projectile on level ground (the level of the cliff top), whose speed we know from energy conservation, and the familiar maximum range u^2/g does the rest.

Solution

The model

Treat the wallet as a particle moving freely under gravity, with no air resistance, thrown from ground level (Padma's height is negligible). Let E be the top edge of the cliff and P the point where I stand, so P is at height h and a horizontal distance d beyond E. The cliff face is vertical, and the beach lies on the far side of it from me, so wherever Padma stands, the wallet has to pass over E (or at worst just graze it) and then come down to P.



We use one fact throughout. If a particle at some level has speed u and moves at angle α above the horizontal, it returns to that level after time $2u\sin\alpha/g$, having travelled horizontally

$$\frac{2u^2\sin\alpha\cos\alpha}{g} = \frac{u^2\sin 2\alpha}{g} \leq \frac{u^2}{g},$$

with equality exactly when $\alpha = 45^\circ$.

The condition is necessary

Suppose some throw gets the wallet to P. It must reach height h, so let Q be the first point of the path at height h. Taking the cliff foot as the zero of potential energy, conservation of energy gives the speed u at Q:

$$u^2 = V^2 - 2gh.$$

Ask yourself Why must Q be at least a horizontal distance d from P?

When the wallet's horizontal position is that of the cliff face, it must be at height at least h, or it would hit the face (or pass below the edge). Its height rises continuously from 0, so it has already reached height h by then: Q is on the beach side of E, or at E. As P is a horizontal distance d beyond E, the horizontal distance from Q to P is at least d.

Between Q and P the wallet goes from height h back to height h, starting with speed u. By the range fact, the horizontal distance it covers is at most u^2/g. Therefore

$$d \leq \frac{u^2}{g} = \frac{V^2 - 2gh}{g}, \qquad \text{that is,} \qquad V^2 \geq g(2h + d).$$

The condition is sufficient

Now suppose $V^2 \geq g(2h + d)$. Then $u^2 = V^2 - 2gh \geq gd > 0$, so $gd/u^2 \leq 1$ and we can choose an angle α with $0 < \alpha \leq 45^\circ$ and

$$\sin 2\alpha = \frac{gd}{u^2}.$$

Consider the parabolic path that passes through E with speed u, moving away from the beach at angle α above the horizontal. By the range fact, it comes back down to height h a horizontal distance $u^2\sin 2\alpha/g = d$ beyond E, which is exactly P. Between E and P it is above height h, so it clears the cliff top.

Run this path backwards in time from E: the wallet is lower and further out over the beach at earlier times (its vertical velocity was larger and upward), so the path meets the ground at some point of the beach, a positive horizontal distance from the cliff foot. Before reaching E the path is entirely on the beach side of the cliff face, so nothing is in the way, and since the beach is broad Padma can stand at that point. By energy conservation the speed there is $\sqrt{u^2 + 2gh} = V$. So if Padma throws from that point at speed V, in the direction of the path, the wallet follows the path and reaches me.

(When $V^2 = g(2h + d)$ exactly, $\alpha = 45^\circ$ and the wallet just grazes the edge E; for larger V there are two angles in $(0^\circ, 90^\circ)$ that work, and the wallet passes strictly above the edge for both.)

Combining the two halves, Padma can throw the wallet to me if and only if $V^2 \geq g(2h + d)$, as required.

Checks

- If $d = 0$ the condition becomes $V^2 \geq 2gh$, which is just what is needed for a vertical throw to reach height h.
- If $h = 0$ it becomes $V^2 \geq gd$, the usual maximum range on level ground.
- Ignoring the cliff edge, the best Padma could do is stand at the cliff foot, and the bounding parabola of all trajectories gives only $V^2 \geq g\left(h + \sqrt{h^2 + d^2}\right)$, which is weaker since $\sqrt{h^2 + d^2} \leq h + d$. So the edge really is the binding constraint.
- Numerically, with $g = 9.8$, $h = 20$ and $d = 15$, the critical speed is $\sqrt{539} \approx 23.2\ \mathrm{m\,s^{-1}}$; Padma stands about 11.4 m from the cliff foot, and a step-by-step simulation of the throw lands it 15.0 m beyond the edge, grazing E. A scan over all launch angles for random h, V confirmed that the largest horizontal gap between the two height-h points of the path is $(V^2 - 2gh)/g$.

Common slips

- Using only the bounding parabola for the point P from the cliff foot. This ignores the cliff edge and gives a condition that is necessary but not sufficient.
- Proving only one direction. The "if" needs an explicit throw, including a check that it starts on the beach and doesn't hit the cliff.

Beyond STEP

The trick of restarting the motion at height h works because the equations of motion don't change with time or under horizontal shifts, so any point of a trajectory can be treated as a fresh launch. Problems of the form "which targets can be reached?" are about the envelope of a family of curves, and with an obstacle in the way the answer is decided by whichever constraint binds first, an idea that reappears in optimisation and the calculus of variations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile motion and energy methods.

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